Lab 4: TCP and UDP Socket Applications

1. Observations Summary

Activity	Command / Input	Local Endpoint	Target / Server	Result / Reply	Note
TCP Exchange	`py echo.py tcp client "hello class"`	`('127.0.0.1', 49848)`	`('127.0.0.1', 12000)`	`Reply: hello class`	Ephemeral client port assigned.
TCP Message Change	`py echo.py tcp client "Mixed CASE 123"`	`('127.0.0.1', 49864)`	`('127.0.0.1', 12000)`	`Reply: Mixed CASE 123`	New ephemeral port assigned by OS.
TCP Stopped Server	`py echo.py tcp client "Testing after ctrl + c server"`	`('127.0.0.1', 49866)`	`('127.0.0.1', 12000)`	`TimeoutError: timed out`	Connection could not be established; timed out after 2s.
UDP Exchange	`py echo.py udp client "hello class"`	`('0.0.0.0', 54378)`	`('127.0.0.1', 12000)`	`Reply: hello class`	Bound to wildcard address with ephemeral port.
UDP Absent Server	`py echo.py udp client "hello class"`	`('0.0.0.0', 54691)`	`('127.0.0.1', 12000)`	`ConnectionResetError: [WinError 10054]`	Windows loopback ICMP port unreachable notification delivered to socket.

2. TCP Message Framing

TCP is a stream-oriented protocol with no built-in message boundaries; bytes sent in multiple calls may arrive together or fragmented across multiple `recv()` calls. To reconstruct discrete messages, a receive loop reads chunks of data until it detects an agreed delimiter. In `echo.py`, the newline character (`\n`) acts as the message framing boundary, indicating that the full message has arrived.

3. Transport Design

- **Fictional Application:** Real-time multiplayer game player-position telemetry.
- **User Requirements:** Low latency is critical. Recent player coordinates must arrive immediately; old coordinates are useless if delayed.
- **Transport Choice:** UDP is preferred because TCP's retransmissions and head-of-line blocking cause lag spikes.
- **Additions Needed over UDP:** The application must add sequence numbers (to discard out-of-order/stale position updates) and lightweight checksums/integrity checks.

4. Reflection

1. **Why is an IP address alone insufficient to identify the service?** An IP address only identifies a host on the network. Port numbers are required to multiplex and route traffic to a specific process or service running on that host.

2. **What supplies TCP message framing here?** The application layer supplies framing by appending a newline delimiter (`\n`) to each message and reading the byte stream in a loop until `\n` is encountered.

3. **Can a UDP application add acknowledgements/retries?** Yes. Reliability can be implemented at the application layer by tracking packet sequence numbers, returning acknowledgement packets, and retransmitting lost packets on timeout.

4. **Does TCP alone supply encryption or a minimum speed?** No. TCP provides reliable, ordered byte delivery, but it transmits data in plaintext without encryption and does not guarantee bandwidth or a minimum transfer speed.